Image Domain Effects on Similarity of Neural Representations

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Abstract

Biological visual systems have evolved around the efficient encoding of natural image statistics to allow animals to successfully recognize complex visual patterns. Recent work has shown that deep neural networks are able to learn similar representations of images as those measured in the visual cortex of animals. This suggests that DNNs may serve as good models for biological visual systems. A question of interest has been how to design and train models to increase their biological similarity to real visual systems. This was explored recently by training neural network models across a range of different data domains, where each domain consists of different natural image statistics. Results from this prior work showed evidence of a positive relationship between the naturalness of the images in different data domains and the degree of biological similarity of the representations learned by the models. Here, we perform a causal analysis on this relationship to better understand how we might train more biologically similar models. Using the network activations from the different models trained on separate data domains, we analyze the causal relationships between training data domain, naturalness of the domain, and network neural predictivity. We find that both the training data domain and the naturalness score of the training data can affect the neural predictivity, and that in fact the causal relationships depend on the specific neural network layer being evaluated and the brain region used for evaluating predictivity.

1 Introduction

The ventral visual stream is a hierarchy of visual processing areas in the brain which helps animals perform the difficult task of visual object recognition. The ventral stream helps animals to quickly identify objects despite large variations to the objects appearance, primarily using a cascade of mainly feed forward computations that culminate in a neural representation of the object [5, 4]. Recently, there has been extensive work that has focused on modeling the neural responses of visual systems using hierarchical deep convolutional neural networks (CNNs). Prior work has produced highly accurate CNN models which can successfully predict the neural responses of primate [18, 17] and mouse [1, 2, 12] ventral visual streams. However, many of these studies simply show correlational relationships between different parts of neural network models and their contribution to their biological similarity. Here, we attempt to perform a causal analysis on work done in [10] which showed that training on natural image statistics is positively correlated with neural predictivity. We use a greedy equivalence search as well as the PC algorithm to test whether there is any actual causal relationship. We show that the training domain and in turn the naturalness score do cause an increase in predictivity. However, we find that this is really only the case in earlier layers of the deep network. This could be due to the fact that the earlier layers of deep CNNs have been shown to be low level feature extractors which may be more sensitive to changes in the underlying image statistics.

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2 Related Work

While there has been extensive work done modeling ventral visual streams, we focus primarily on work exploring the effects of different training regimes and network architecture decisions. In [11], Schrimpf et al. looked specifically at which CNN architecture was actually the most biologically similar to primate visual ventral streams. They proposed a method called Brain-Score which scored different neural networks on how brain-like they were based on simple object recognition. The assumption was that the closer the model architecture is to a real brain, the better performing the model would be. The results indicated that complexity of models was only important to a degree, and that a more brain-like architecture was not necessarily indicative of better neural predictivity.

There has also been considerable work focused on modelling the mouse visual ventral stream, where good performance has proved more elusive than for primates. Similar to [11], Conwell et al. looked at various different model architectures to see if mouse visual streams could indeed be better modelled [2]. The research specifically looked at model architectures outside of traditional convolutional neural networks, finding that combinations of different architectural designs could help improve neural predictivity of the mouse ventral stream. Somewhat contrary to this finding, Conwell et al. compared 85 different deep neural network architectures trained on human fMRI data of 16 parts of the brain visual system and found little effect of model architecture on neural predictivity. However, changes to the learning task showed that self-supervised models showed relatively stronger neural predictivity across the board.

In [13], Shi et al. note that mouse visual stream is less hierarchical than primate visual streams found in primates. This begs the question whether feed forward convolutional nueral networks are actually the best architecture to model mouse ventral systems. They introduce a CNN model structure that is biologically constrained to learn the lateral areas of the mouse ventral visual stream, thus offering further architectural choices for modelling the ventral system.

Many of the developed methods for modeling the ventral stream has focused on using supervised deep neural network architectures. However, a common issue with this framework is the need for more labelled training data when it is not available. Zhuang et al. brought developments from unsupervised learning to build neural network models with deep unsupervised constrastive embedding methods. They found that self-supervised learning models achieve equal or greater performance in multiple primate ventral visual areas. This has led to a larger acceptance of self-supervised methods which are able to take advantage of the larger amounts of unlabelled training data. Work by Nayebi et al. has reaffirmed this point when looking at the performance of unsupervised constrastive embedding methods when modeling the mouse visual system [8]. They found that using task-agnostic objective functions were substantially better than models trained with supervised objectives.

Recent research has primarily focused on understanding the effects that different model architectures can have on neural predictivity of the ventral stream. As shown, there has been mixed results, with some model architectures proving better than others, but there has not appeared a consistent consensus outside of using some form of deep neural networks and self-supervised learning. There has been relatively little work that has focused on the training data as a cause of better neural predictivity. In the work of Nayebi et al. [8] the authors did uncover that models trained on low resolution images performed the best, perhaps due to the ecological niche under which that mice visual systems evolved. As mice, and all animals, develop their visual systems under a specific ecological and biological setting where the images seen can vary dramatically, the data that models are trained on could be a key factor in improving neural predictivity. This is the premise of research done by one of our authors in [10], where separate models were trained on images from different datasets, where each dataset consisted of images with a different degree of naturalness. Naturalness simply measures literally how natural an image appears compared with nature. The author found that there appeared to be a positive association between the degree of naturalness and the neural predictivity of the visual stream models. We specifically extend this work to find the causal relationship between neural predictivity, training domain naturalness, and the final neural predictivity of a model. Understanding the causal relationship can further inform whether the naturalness of images in the training dataset needs to be considered when building a model of mice visual systems and possibly for primates as well.

3 Methods

We use data from Prasad, Manor, and Pereira [10], and describe below in Subsection 3.1 the initial datasets used in [10] and in Subsection 3.2 the methods used by [10] to generate the dataset. Finally, in Subsection 3.3 we describe the methods of causal inference we employ in our experiments, described in Section 4

3.1 Datasets

Figure 1: Sample images from each dataset. Networks were trained on ImageNet, DomainNet, and CREMI, then evaluated on AIVC and Sensorium.

ImageNet [3] is a multiclass object categorization dataset with $C = 1000$ classes containing a training set with around 1.3 million images and a validation set of around 50 thousand images. In order to control for dataset size and since the dataset is already downsized to 64×64 (shown to be important for modeling mouse brains [8], we used the TinyImageNet [7] dataset which contains about 100k training images and about 20k validation images with only 200 classes.

DomainNet [9] is a large-scale multi-source domain adaptation dataset containing 6 domains each having 345 categories: Real (510 images per category/175,327 total), Infograph (150 images per category/53,201 total), Clipart (150 images per category/48,833 total), Painting (220 images per category/75,759 total), Sketch (220 images per category/70,386 total), and Quickdraw (500 images per category/17,2500 total).

CREMI(**C**ircuit **R**econstruction from **E**lectron **M**icroscopy **I**mages) is an electron microscopy dataset made up of three datasets, each consisting of two 5 μm^3 volumes (training and testing, each $1250 \times 1250 \times 125$ px of serial section EM of the adult fly brain)[1]. We treated each slice in the z dimension as a separate image and then generated 100 125×125 tiles per image resulting in a dataset with 27,750 images which we then trained on.

Sensorium is a large-scale dataset from mouse primary visual cortex containing the responses of more than 28,000 neurons across seven mice stimulated with thousands of natural images, together with simultaneous behavioral measurements that include running speed, pupil dilation, and eye movements [16].

Allen Institute Visual Coding Dataset (**AIVC**) [15, 14] is a mouse neural recording dataset containing electrophysiological and calcium fluorescence recordings of around 60 thousand neurons from the VISp, VISl, VISal, VISam, VISpm, and VISrl regions of 221 different mice. We limited our study to the neural responses of animals in response to the Natural Scenes stimulus set which consisted of 118 grayscale natural images. After following the filtering process outlined by [2] and [8] we were left with around 17,000 neurons across 98 different specimen(calcium) and 1422 neurons across 42 different specimen(neuropixels) across the six visual areas.

3.2 Data

The data for this project comes from work done in [10], where several different self-supervised CNN models were trained on datasets from different domains. We list the key datasets above and note that a model was trained on ImageNet [3], CREMI, and six different domains of DomainNet [9], specifically, real, painting, clipart, infograph, sketch, and quickdraw, as shown in Figure 1. Our work then uses the activations extracted from the models at each layer for each model. We then perform our causal analysis on the activations for each layer with respect to the naturalness of the image domain and the neural predictivity of the model. We calculate naturalness by sampling 1000

[1] MICCAI Challenge on Circuit Reconstruction from Electron Microscopy Images (CREMI): `https://cremi.org`

images from each and calculating the power spectrum $1/f^p$ of each image, then comparing to the idealized $p = 2$ by taking $-\log|p-2|$, then averaging over the sample. The neural predictivity of each model is computed over 100 trials using an inter-animal consistency metric defined in [8] with mouse neural data from Sensorium [16] and AIVC [14, 15]. In particular, we use the neural predictivity from the AIVC calcium dataset.

3.3 Causal Analysis

Since we already have the data from [10], we can use a causal discovery procedure to build a structural causal model on the variables we wish to investigate. Because the neural network architecture and training method is fixed, we do not expect the presence of latent confounders or causal feedback. Thus we can use a constraint-based method like the PC algorithm. We note, though, that training domain, as well as other variables in the data like network layer and evaluation region, are categorical rather than continuous variable. This means that we will need to use a kernel-based independence test in the causal analysis in order to apply it to our mixed data. We can also use a score-based method like GES, which supports mixed data with the use of a generalized score [6].

Using a constraint-based or score-based causal discovery method, we would like to be able to discriminate between two potential causal models which we illustrate in Figure 2. In either model, we can be confident of our edge orientations (since there is no way that, for example, the neural predictivity measured in this experiment would cause an image to be from a different training domain). Thus the question is primarily that of the underlying undirected graphical model, and we describe in the next section our findings.

Figure 2: Candidate structural causal models with naturalness (X), neural predictivity (Y), and image domain (Z). The image domain Z may be a confounder for the effect of X on Y (left), or affect only X directly (right).

4 Experiments

We begin by performing the PC algorithm using a Gaussian kernel conditional independence test at significance level $p < 0.05$. We use a kernel independence test because our data is mixed, and the results on a random sample of size $n = 1000$ are shown in Figure 3. (We subsample our data because using kernel independence tests for the full $n = 27000$ data points proved intractable.) We see the layer and evaluation region both affect the neural predictivity, but the PC algorithm output indicates that there is no causal relationship between the naturalness and the predictivity. Note that we can orient the unoriented edge between domain and naturalness in the causal graph using domain knowledge, since we know the underlying process that generates the image domain. We then perform GES the same sample using the generalized cross-validation score [6], again because of its suitability for mixed data (and again using a subsample due to runtime concerns. The output is shown in Figure 4. The GES output shows a causal relationship between naturalness and predictivity, with no direct causal relationship between domain and predictivity.

To investigate this discrepancy between the PC and GES output, we restrict our attention to a single layer and region at a time and use GES on each subset, with results in 5. Note that we can direct the undirected edges with domain knowledge. Following [10], we note that shallower layers (particularly conv1) show more neural predictivity, and we see that deeper layers (particularly conv4, conv5) show v-structures which we know to be spurious based on domain knowledge. Examining shallower layers, however, we see that the causal relationships of domain and naturalness on predictivity actually depend on the evaluation region. For example, we see that the predictivity of layer conv1 with respect to the VISp region depends on both image domain and naturalness, but with respect to VISal depends only on naturalness.

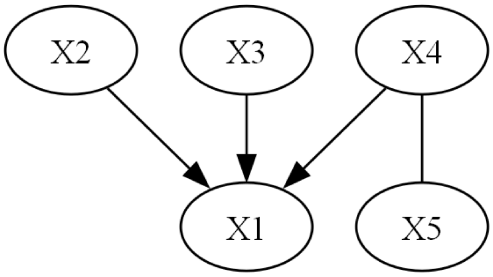

Figure 3: PC algorithm result on a sample of neural predictivity data ($n = 1000$). Variable $X1$ is the neural predictivity score, $X2$ the layer, $X3$ the evaluation region, $X4$ the training domain, and $X5$ the naturalness score.

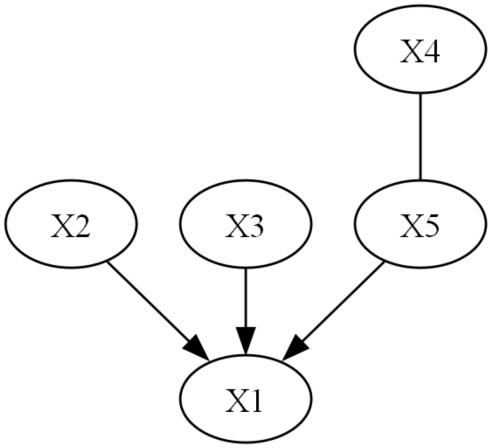

Figure 4: GES output on a sample of neural predictivity data ($n = 1000$). Variable $X1$ is the neural predictivity score, $X2$ the layer, $X3$ the evaluation region, $X4$ the training domain, and $X5$ the naturalness score.

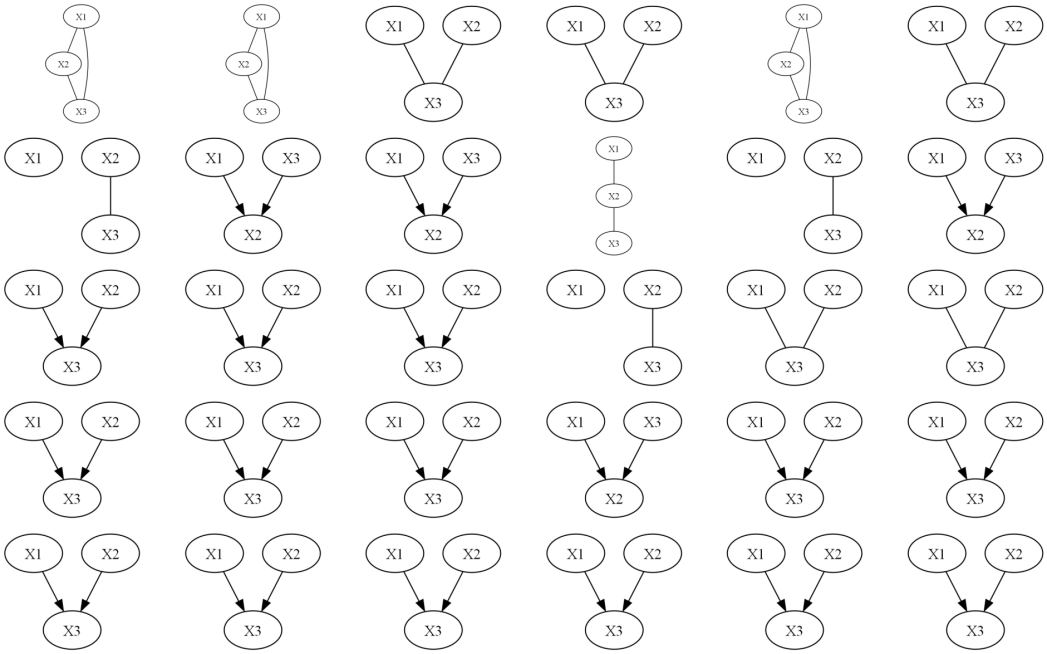

Figure 5: GES output on a sample of neural predictivity data ($n = 1000$). Variable $X1$ is the neural predictivity score, $X2$ the training domain, and $X3$ the naturalness score. Each column represents one brain region (from left to right: VISp, VISl, VISal, VISam, VISpm, VISrl) and each row is one layer in the AlexNet (from top to bottom: conv1, conv2, conv3, conv4, conv5).

5 Conclusion

We have seen that both the training image domain and the training data's naturalness can affect the neural predictivity of a trained convolutional neural network, and further that this effect may actually depend on the layer of the network being evaluated and the brain region being used for predictivity evaluation. In other words, the layer and region variables not only affect the predictivity, but also the causal structure itself. Further clarifying these relationships, then, might focus on the predictivity of neural network layers for each specific brain region. One could also engineer training datasets which are drawn from different image domains but that have controlled naturalness scores in order to investigate the causal relationship between naturalness and predictivity, or training on subsets of each image domain that have the same naturalness to investigate the causal relationship between domain and predictivity. We leave these possibilities to future work.

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